

Renewable energy

Fossil fuels such as oil and gas power our world, but these resources are not unlimited. Their continued use also comes with serious environmental problems, including air pollution. As a result, we are increasingly turning to wind, water, and the sun in search of more sustainable and less destructive sources of energy.

Water power

- **What?** Hydroelectricity
- **Who?** William Armstrong
- **Where and when?** UK, 1878

While fishing in the UK, William Armstrong was watching a waterwheel and it occurred to him that it harnessed only a small amount of power from the water. Armstrong dammed a nearby river to create a lake, making his house the first in the world to be powered by a hydroelectric generator.



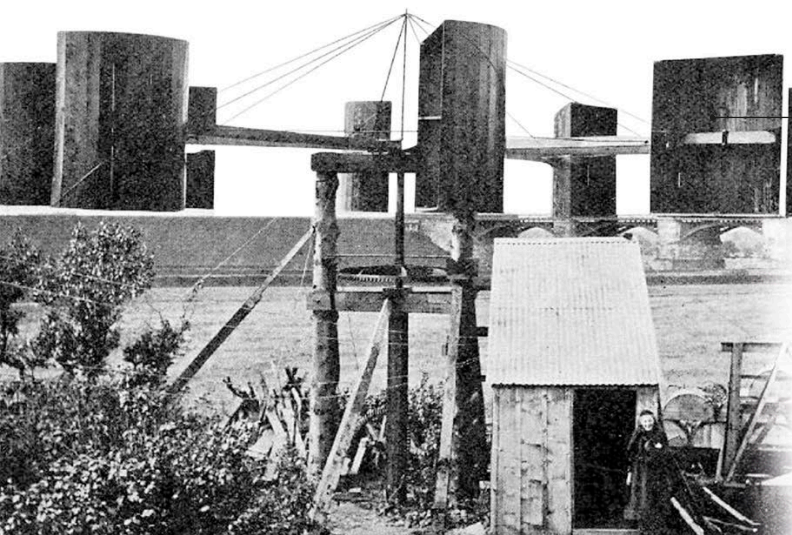
Hoover Dam, built in 1936, is a source of hydroelectric power

Wind power

- **What?** Wind-powered electric generator
- **Who?** James Blyth
- **Where and when?** Scotland, 1887

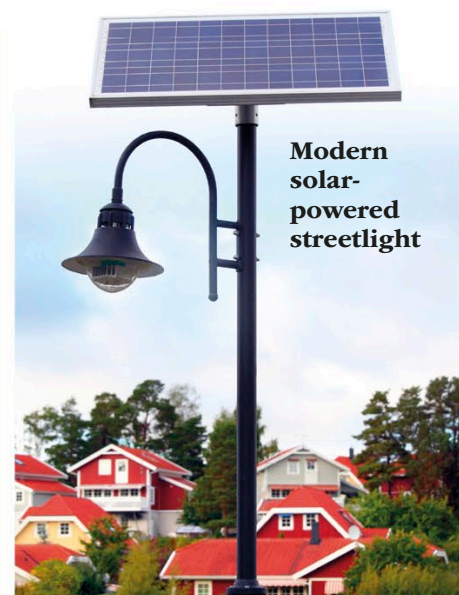
After constructing a windmill, James Blyth attached to it an electric motor to light his

cottage. He offered to light his village's main street, but locals thought the strange light was the work of the devil. He did, however, build a larger wind generator to provide electricity for a hospital in a nearby town.



Blyth's windmill had horizontal sails rather than vertical.

Blyth's electricity-generating windmill



Modern solar-powered streetlight

Solar power

- **What?** Solar-powered electricity
- **Who?** Augustin Mouchot
- **Where and when?** France, 1869

Math teacher Augustin Mouchot was convinced that coal would eventually run out. In 1860, he began experimenting with capturing the heat from the sun. In 1869 he displayed a "solar steam engine" in Paris. Unfortunately, coal remained cheap and plentiful and Mouchot's work was ignored.

Geothermal plant, Iceland



Electricity from hot springs

- **What?** Geothermal electric generator
- **Who?** Piero Ginori Conti
- **Where and when?** Italy, 1904

The Romans used hot springs to heat their buildings, as people have in Boise, Idaho, since 1892. However, Piero Ginori Conti was the first person to demonstrate a geothermal power generator, in 1904 in Larderello, Italy. The first commercial geothermal power station was built there in 1911.